AudioConditioner

Results & Applications

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CSC_52002_EP - Multimodal generative AI

APPLICATION I: IMAGE TO AMBIENT MUSIC

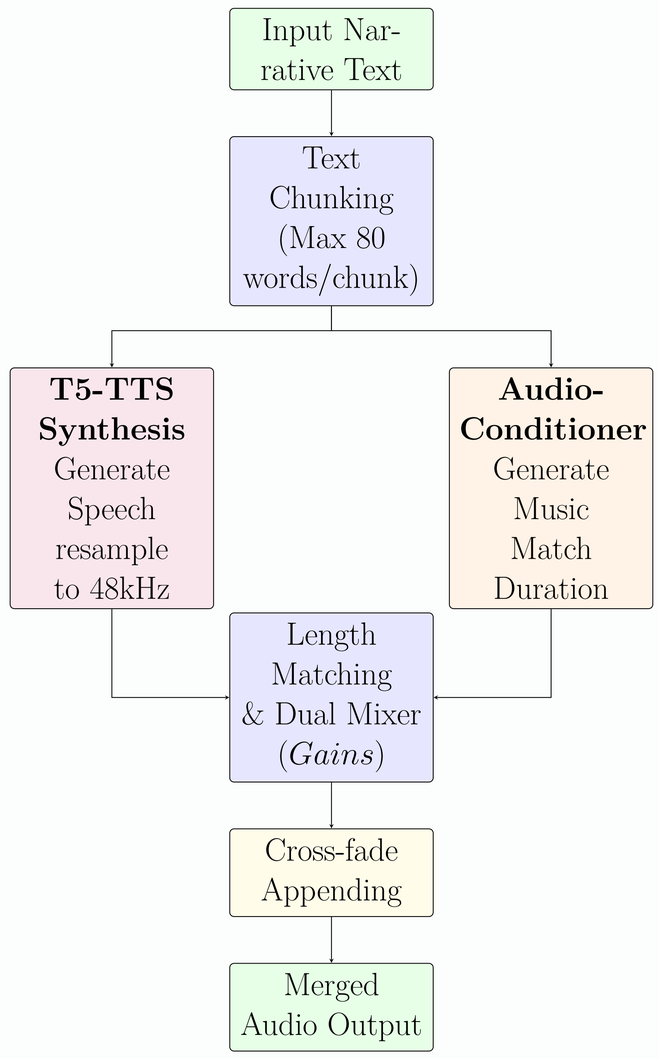
One direct and powerful application of the **AudioConditioner** framework is generating ambient music and atmospheres based solely on visual inputs. By passing an image into our pipeline:

- The **BLIP Model** automatically captions the image's scene, extracting emotional elements, colors, objects, and implicit narrative.
- This scene description drives the **AudioConditioner** pipeline (through the **CLAP Descriptor** and **Stable Audio**), outputting high-fidelity audio that perfectly encapsulates the mood of the image.
- The audio candidates are re-ranked using the **CLAP Cosine Similarity Filter** to ensure the highest alignment with the initial visual.

This enables seamless soundtrack generation for static artwork, game environments, or visual media.

APPLICATION II: TEXT READING WITH ADAPTIVE MUSIC

The **ReadMusic** module merges Speech Synthesis with adaptive background music scoring, turning textual narratives into fully immersive audiobooks.



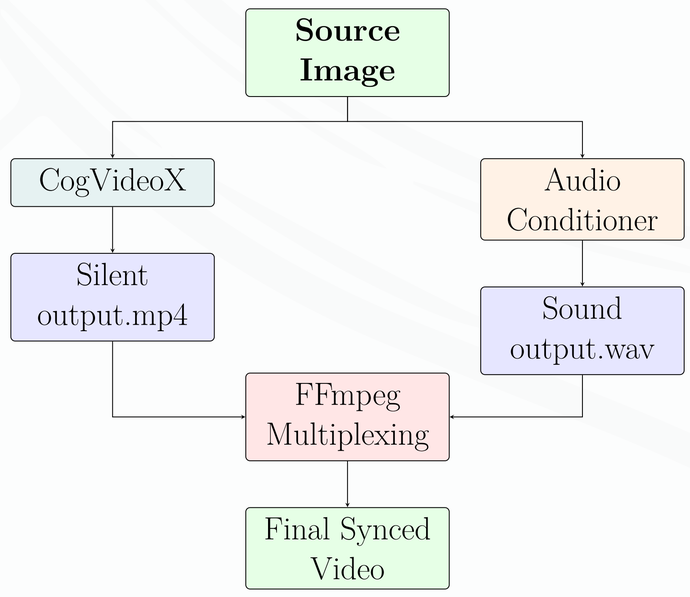
The text is robustly split using natural pauses (punctuation, newlines). For each chunk (mandated due to the strict **47-second generation maximum** of the under-the-hood Stable Audio model), the script runs the T5 text-to-speech engine concurrently with our Music Generation model. Finally, speech and music arrays are zero-padded to match lengths, multiplied by their specific gains, and blended with dynamic crossfades to avoid clipping artifacts.

APPLICATION III: VIDEO GENERATION & ORCHESTRATION

Taking the visual modality further, the `video_sound.py` application creates animated cinematic clips with a tightly synchronized and responsive musical score from a single still image.

Parallel Execution Pathway:

1. **Image Animation:** The input image is parsed through the **CogVideoX** Model. As we strictly utilized an *open-weight model* for this task, the resulting video quality remains relatively limited but serves as a functional proof-of-concept for the pipeline.
2. **Image Orchestration:** Simultaneously, the image goes directly into the **AudioConditioner** model to extract visual semantics and produce a matching audio waveform.
3. **Multimodal Merging:** We integrate **FFmpeg** through subprocesses to automatically compile the generated silent video (`.mp4`) and the generated audio (`.wav`), multiplexing them into a single presentation-ready file.



These applications showcase the great versatility of **AudioConditioner**, positioning it as a strong foundation for automated scoring, procedural art generation, and immersive multimodal storytelling.
However, due to the limitation to open-weight models, the quality of the generated video is not as good as the quality of the generated audio. Many artifacts are often visible in the generated video.

IF WE HAD MORE TIME

Given more time and compute budget, we would have fine-tuned our model using **Proximal Policy Optimization (PPO)**. By using the CLAP similarity score (between the generated audio and the text prompt) as a reward signal, the model could strictly optimize itself to directly maximize alignment instead of just distilling a Teacher LLM.

- **The Bottleneck:** In RL algorithms like PPO, the agent needs to continually generate outputs (rollouts) to explore the environment and receive rewards.
- **Order of Magnitude:** A single audio generation step with **Stable Audio** takes approximately **1 minute** to complete.
- **Conclusion:** Gathering enough generation trajectories to meaningfully update the network via PPO would thus require **weeks of continuous computing time**, making it unfeasible within the scope of this project.

REFERENCES: MODELS USED

- **BLIP (Bootstrapping Language-Image Pre-training):** Visual scene semantic extraction and captioning.
- **CLAP (Contrastive Language-Audio Pretraining):** Text semantic embeddings and cosine similarity candidate re-ranking.
- **Stable Audio (Open):** Core diffusion model for structural 47-second music sequence generation.
- **T5-TTS:** Robust text-to-speech synthesis engine utilized in the ReadMusic narrative pipeline.
- **CogVideoX:** Advanced transformer-based text-to-video diffusion model for high-consistency visual animation.

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